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Real-time SQL Performance Tuning & Optimization interview questions:

Query tuning, indexing, execution plans, statistics, locking, blocking, waits, partitioning, tempdb, hardware, monitoring, HADR performance.

General Performance Tuning Concepts

1. What is SQL Server performance tuning and why is it important?
2. What are the first steps you take when investigating a slow-running query?
3. How do you identify whether performance issues are at the SQL level or infrastructure level?
4. What is the difference between proactive and reactive performance tuning?
5. How do you baseline SQL Server performance?
6. Which DMV would you use to check recent expensive queries?
7. What's the difference between CPU-bound, IO-bound, and memory-bound performance issues?
8. What is parameter sniffing in SQL Server?
9. How do you detect parameter sniffing problems?
10. What's the difference between recompilation and parameterization?
11. How do you identify query regressions after an upgrade?
12. What are the top performance tuning tools you use?
13. How do you capture a SQL Server query workload for analysis?
14. What's the role of the Query Store in SQL Server?
15. What's the difference between estimated vs actual execution plans?
16. How do you compare execution plans between two versions of SQL Server?
17. What are common causes of slow queries after deployment?
18. How do you measure query performance in SQL Server?
19. What is the role of statistics in performance?
20. How do you handle performance regressions caused by changes in statistics?

Index Tuning

21. What are clustered indexes?
22. What are non-clustered indexes?
23. What's a covering index?
24. What's a filtered index?
25. How do you identify missing indexes?
26. What's the risk of over-indexing a table?
27. How do you detect unused indexes?
28. How does index fragmentation affect performance?

29. How do you rebuild vs reorganize indexes?
30. What's the impact of fill factor on index performance?
31. How do you decide which columns to include in a non-clustered index?
32. What are indexed views and when to use them?
33. How do you troubleshoot queries that are not using indexes?
34. What is index seek vs index scan vs table scan?
35. How does SQL Server choose between seek and scan?
36. What is a heap table and performance implications?
37. When should you use composite indexes?
38. How do you detect duplicate indexes?
39. What is an index hint and when should you use it?
40. How do you manage indexes on very large tables?

Execution Plans

41. What is an execution plan?
42. How do you generate an execution plan in SQL Server Management Studio?
43. What are common operators in execution plans?
44. How do you read a nested loop join operator?
45. What is the difference between hash join and merge join?
46. What is key lookup in execution plan?
47. How do you eliminate key lookups?
48. What is a bookmark lookup?
49. What is a spool operator?
50. What does "Estimated number of rows" vs "Actual number of rows" indicate?
51. How do you detect cardinality estimation issues?
52. What is a parallelism operator in execution plans?
53. How do you control parallelism?
54. What is MAXDOP?
55. How do you identify execution plan regressions?
56. What are trivial vs full execution plans?
57. How do query hints affect execution plans?
58. What is forced plan in Query Store?
59. What is plan cache?
60. How do you clear the plan cache safely?

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Statistics

61. What are statistics in SQL Server?
62. How do statistics improve query performance?
63. How do you update statistics manually?
64. What is auto-create statistics?
65. What is auto-update statistics?
66. When does auto-update statistics trigger?
67. How do you detect outdated statistics?
68. What's the difference between fullscan and sample statistics?
69. How do statistics affect cardinality estimation?
70. What is the cardinality estimator in SQL Server?
71. How did the cardinality estimator change in SQL Server 2014?
72. How do you troubleshoot CE-related regressions?
73. How do you enable legacy cardinality estimator?
74. What is histogram in statistics?
75. How do filtered statistics work?
76. What's the role of ascending key problem in statistics?
77. How do statistics impact query plans in partitioned tables?
78. How do you monitor statistics updates in SQL Agent jobs?
79. How do statistics affect parameter sniffing?
80. What's the impact of auto-update statistics async option?

Query Tuning

81. How do you rewrite a slow SELECT query?
82. What is SARGability?
83. How do you detect non-SARGable predicates?
84. What is an example of a non-SARGable query condition?
85. Why should you avoid functions in WHERE clauses?
86. How do you reduce sorting in queries?
87. How do you optimize queries using EXISTS vs IN?
88. What is the difference between NOT IN and NOT EXISTS performance?
89. How do you use APPLY for performance optimization?
90. How do you decide between CTE and temp tables?
91. What is the performance impact of table variables?
92. What is the performance difference between UNION and UNION ALL?

93. How do you optimize pagination queries?
94. How do you avoid correlated subqueries?
95. What's the impact of DISTINCT in queries?
96. How do you optimize GROUP BY queries?
97. How do window functions affect performance?
98. How do you tune dynamic SQL queries?
99. What is the performance impact of scalar UDFs?
100. How do inline table-valued functions improve performance?

Locks, Blocks & Concurrency

101. What is a deadlock?
102. How do you capture deadlock graphs?
103. How do you avoid deadlocks?
104. What is lock escalation?
105. What is blocking in SQL Server?
106. How do you identify blocking sessions?
107. How do you troubleshoot long blocking chains?
108. What is optimistic vs pessimistic concurrency?
109. How do snapshot isolation levels affect performance?
110. What is READ COMMITTED SNAPSHOT isolation?
111. What is the cost of using SERIALIZABLE isolation?
112. How do you identify lock waits in DMVs?
113. What are common wait types for locking issues?
114. How do you minimize lock contention?
115. What is NOLOCK hint and its risks?
116. What is row versioning in SQL Server?
117. How do you detect latch contention?
118. How do you troubleshoot TempDB contention?
119. How do you identify excessive recompiles?
120. How do triggers impact performance?

Wait Statistics & Monitoring

121. What is wait statistics analysis?
122. What DMV shows top wait types?
123. What are CXPACKET and CXCONSUMER waits?

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124. How do you resolve high PAGEIOLATCH waits?
125. What are SOS_SCHEDULER_YIELD waits?
126. How do you monitor query execution time by DMV?
127. How do you capture long-running queries?
128. What is Extended Events and how do you use it?
129. What's the difference between Profiler and Extended Events?
130. What is PerfMon and what counters matter for SQL?
131. What is the role of Activity Monitor?
132. How do you capture blocking sessions in real time?
133. What is Query Store's role in wait analysis?
134. What is Live Query Statistics?
135. How do you measure batch request/sec?
136. What is the threshold for acceptable CPU usage in SQL Server?
137. How do you detect memory pressure?
138. What's the difference between buffer cache hit ratio and page life expectancy?
139. How do you troubleshoot low PLE?
140. What is resource governor?

TempDB & Memory Optimization

141. Why is TempDB important for performance?
142. How do you configure TempDB optimally?
143. What is the recommended number of TempDB files?
144. What is TempDB contention?
145. How do you detect TempDB contention?
146. How do you fix PAGELATCH_UP waits?
147. What is the impact of version store on TempDB?
148. How do you clean up version store?
149. What is memory grant feedback?
150. How do you monitor memory grants?
151. What is buffer pool extension?
152. How do you reduce spills to TempDB?
153. What is memory-optimized table variable?
154. How do you decide memory allocation for SQL vs OS?
155. What is In-Memory OLTP?
156. How do memory-optimized tables improve performance?

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157. What are limitations of In-Memory OLTP?
158. What is native compilation?
159. How do you monitor memory clerks?
160. How do you troubleshoot RESOURCE_SEMAPHORE waits?

Partitioning & Large Tables

161. What is table partitioning?
162. How does partitioning help performance?
163. What is partition elimination?
164. How do you check if a query is using partition elimination?
165. What are sliding window scenarios in partitioning?
166. What is partition scheme and partition function?
167. How do you switch partitions in/out?
168. How does partitioning affect statistics?
169. What's the difference between partitioned views and table partitioning?
170. How do you detect skewed partitions?
171. How do you design partition keys?
172. What is indexed partitioned table limitation?
173. How does partitioning impact maintenance jobs?
174. What is parallelism with partitioning?
175. How do you troubleshoot partitioned table performance issues?
176. What is partition alignment?
177. How do you rebuild indexes on partitioned tables efficiently?
178. How do you back up partitioned tables?
179. How does compression help large tables?
180. What is columnstore index and its use case?

Columnstore Indexes

181. What is a columnstore index?
182. What's the difference between clustered and non-clustered columnstore indexes?
183. How do columnstore indexes improve performance?
184. What is segment elimination?
185. What is batch mode execution?
186. How do you troubleshoot columnstore performance issues?
187. How do dictionary and segment affect compression?

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188. What is rowgroup in columnstore?
189. What are delta stores in columnstore?
190. How do you monitor columnstore index health?
191. What are limitations of columnstore indexes?
192. How do you rebuild columnstore indexes?
193. How do you choose between rowstore vs columnstore?
194. How do you handle updates in columnstore indexes?
195. How do columnstore indexes impact memory?
196. What is hybrid columnstore + rowstore design?
197. How do you check if query uses batch mode?
198. What are row mode vs batch mode operators?
199. How do you optimize analytics queries with columnstore?
200. What is archival compression in columnstore?

Query Store & Automatic Tuning

201. What is Query Store?
202. How do you enable Query Store?
203. How do you force a plan in Query Store?
204. How do you detect query regressions with Query Store?
205. What are Query Store capture modes?
206. How do you clear Query Store?
207. How do you monitor top resource-consuming queries in Query Store?
208. How does automatic tuning work in SQL Server?
209. What is plan correction?
210. How do you enable automatic plan correction?
211. How do you check if a plan is forced automatically?
212. What are Query Store wait statistics?
213. How does Query Store help with parameter sniffing?
214. What is query runtime statistics aggregation?
215. How do you monitor forced plan stability?
216. What is stale plan forcing?
217. How do you disable a forced plan?
218. How do you track query performance over time with Query Store?
219. How do you tune Query Store storage settings?
220. How do you troubleshoot Query Store performance overhead?

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Monitoring & DMVs

221. Which DMV shows missing indexes?
222. Which DMV shows index usage stats?
223. Which DMV shows cached execution plans?
224. How do you list top CPU-consuming queries?
225. How do you find top IO-consuming queries?
226. How do you find queries with most execution count?
227. Which DMV shows memory grants?
228. Which DMV shows lock information?
229. Which DMV shows latch contention?
230. Which DMV shows wait stats per query?
231. How do you check plan cache size?
232. How do you detect ad-hoc query pollution in plan cache?
233. How do you check buffer pool usage?
234. How do you check page life expectancy by NUMA node?
235. Which DMV shows missing statistics?
236. Which DMV shows partition stats?
237. Which DMV shows query store forced plans?
238. Which DMV shows spilled sorts to TempDB?
239. Which DMV shows parallelism usage?
240. How do you detect unused indexes in DMVs?

Server & Instance Tuning

241. How do you configure MAXDOP?
242. What is cost threshold for parallelism?
243. How do you monitor worker thread exhaustion?
244. What is lightweight pooling?
245. How do you configure instant file initialization?
246. What is auto-growth impact on performance?
247. How do you pre-size database files?
248. How do you configure multiple data files for TempDB?
249. How do you monitor autogrowth events?
250. What is lock pages in memory?
251. How do you set max server memory in SQL Server?
252. What is minimum memory per query?

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253. How do you detect NUMA node imbalances?
254. How do you configure affinity mask?
255. What is priority boost and should you use it?
256. How do you optimize SQL Server startup parameters?
257. What is trace flag 1117 and 1118?
258. How do you use startup trace flags safely?
259. What is soft-NUMA?
260. How do you tune SQL Server for VM environments?

Query Optimization in Real Scenarios

261. How do you tune queries with multiple joins?
262. How do you optimize a query with OR conditions?
263. How do you improve performance of LIKE queries?
264. How do you optimize full-text search queries?
265. How do you rewrite queries with large IN clauses?
266. How do you deal with skewed data distributions?
267. How do you optimize queries involving CROSS APPLY?
268. How do you tune JSON queries in SQL Server?
269. How do you optimize XML queries in SQL Server?
270. How do you avoid implicit conversions?
271. How do you detect implicit conversions in execution plans?
272. What is the performance impact of implicit conversions?
273. How do you fix queries with arithmetic overflows?
274. How do you deal with large result sets in client apps?
275. How do you tune queries with TOP + ORDER BY?
276. How do you detect parameter-sensitive plans?
277. What are plan guides and when do you use them?
278. How do you optimize queries with windowing functions?
279. How do you tune queries with CASE expressions?
280. How do you detect performance problems caused by scalar UDFs?

Advanced Performance Tuning

281. How do you optimize queries with hierarchical data (CTE recursion)?
282. How do you tune MERGE statements?
283. How do you optimize bulk inserts?

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284. How do you tune data warehouse ETL loads?
285. How do you tune batch operations?
286. How do you optimize cross-database queries?
287. How do you optimize linked server queries?
288. How do you reduce network roundtrips in queries?
289. How do you tune OPENQUERY performance?
290. How do you troubleshoot slow distributed queries?
291. How do you tune service broker performance?
292. How do you tune replication performance?
293. How do you tune Change Data Capture performance?
294. How do you tune AlwaysOn readable secondary performance?
295. How do you tune Log Shipping for latency?
296. How do you monitor latency in Availability Groups?
297. How do you tune AG automatic seeding?
298. How do you troubleshoot AG synchronization performance?
299. How do you detect long redo queue in AG?
300. How do you minimize log send rate bottlenecks?

HADR & Performance

301. How do you monitor performance impact of synchronous AG replicas?
302. What is the difference between synchronous vs asynchronous AG?
303. How do you reduce redo latency in AG?
304. How do you tune AG readable secondary workload?
305. What is preferred replica routing?
306. How do you tune log compression in AG?
307. How do you measure AG failover performance?
308. How do you monitor AG network performance?
309. How do you detect AG bottlenecks with DMVs?
310. What is AG read workload balancing?
311. How do you optimize AG seeding performance?
312. How do you troubleshoot slow AG failover?
313. How do you monitor AG distributed transactions?
314. What are AG wait types affecting performance?
315. How do you optimize log shipping backup frequency?
316. How do you minimize latency in log shipping?

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317. How do you detect backup IO performance issues?
318. How do you tune backup compression performance?
319. How do you minimize restore time for large databases?
320. How do you measure VLF impact on log performance?

Storage & IO

321. How do you detect IO bottlenecks in SQL Server?
322. What PerfMon counters help with IO?
323. What is write-ahead logging?
324. How do you optimize log file placement?
325. How do you optimize data file placement?
326. What is filegroup strategy for performance?
327. How do you optimize tempdb IO?
328. How do you detect and fix parameter sniffing issues in stored procedures?
329. How do you monitor latency on SAN storage?
330. What's the impact of disk queue length on SQL Server performance?
331. How do you measure log write latency?
332. How do you optimize database file autogrowth settings?
333. What's the performance impact of too many VLFs?
334. How do you reduce number of VLFs?
335. What is instant file initialization and why is it important?
336. How do you troubleshoot PAGEIOLATCH waits?
337. What's the role of checkpoint in performance?
338. How do you tune checkpoint frequency?
339. How do you detect stalled checkpoint processes?
340. How do you optimize TempDB for heavy workload?
341. How do you monitor stalled IO requests?
342. What's the difference between read-ahead IO and random IO?
343. How does SQL Server optimize sequential scans?
344. How do you measure file latency with sys.dm_io_virtual_file_stats?
345. How do you handle IO bottlenecks on VMs?
346. How do you troubleshoot write log (WRITELOG) waits?
347. What's the role of indirect checkpoints in SQL Server?
348. How do you configure indirect checkpoints for large databases?

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Parallelism & CPU Optimization

349. What is parallelism in SQL Server?
350. How do you detect parallel query execution in execution plans?
351. What is CXPACKET wait?
352. How do you resolve CXPACKET waits?
353. What is CXCONSUMER wait?
354. What is the impact of MAXDOP on query performance?
355. How do you decide MAXDOP value for OLTP workloads?
356. How do you decide MAXDOP value for DW workloads?
357. What is cost threshold for parallelism?
358. How do you tune cost threshold for parallelism?
359. What is intra-query parallelism?
360. How do you detect parallelism imbalance?
361. What is parallel plan skew?
362. How do you avoid over-parallelization?
363. How do you identify queries that benefit from parallelism?
364. What's the impact of scalar functions on parallelism?
365. How do you troubleshoot SOS_SCHEDULER_YIELD waits?
366. How do you measure CPU utilization per query?
367. How do you detect CPU bottlenecks with DMVs?
368. How do you troubleshoot high compilations per second?

Memory Optimization

369. How does buffer pool work in SQL Server?
370. What is lazy writer process?
371. How do you detect memory pressure?
372. What is stolen memory in SQL Server?
373. How do you monitor memory clerks?
374. What is RESOURCE_SEMAPHORE wait?
375. How do you resolve RESOURCE_SEMAPHORE waits?
376. How do you control query memory grants?
377. What is memory grant feedback?
378. How do you troubleshoot spills to TempDB?
379. What is hash warning event?
380. How do you monitor page life expectancy (PLE)?

381. What is acceptable threshold for PLE?
382. What is buffer cache hit ratio?
383. How do you interpret buffer cache hit ratio?
384. How do you troubleshoot stolen memory growth?
385. What is columnstore memory usage?
386. How do you configure lock pages in memory?
387. What is plan cache bloat?
388. How do you clear plan cache?

Query Design Optimization

389. How do you tune queries with too many joins?
390. How do you optimize subqueries?
391. How do you replace cursors with set-based logic?
392. What's the performance impact of cursors?
393. How do you detect cursor usage in SQL Server?
394. What's the difference between fast_forward and static cursors?
395. How do you optimize queries using EXISTS?
396. What is the difference between EXCEPT and NOT EXISTS?
397. How do you optimize string search queries?
398. How do you optimize LIKE '%string%' queries?
399. What's the impact of wildcards on indexes?
400. How do you tune queries using BETWEEN?
401. How do you tune queries using CASE expressions?
402. How do you optimize GROUP BY with large data?
403. What's the performance impact of HAVING clause?
404. How do you tune TOP N queries?
405. How do you optimize DISTINCT queries?
406. How do you optimize window functions?
407. How do you optimize ranking functions (ROW_NUMBER)?
408. How do you tune CROSS JOIN queries?

Transactions & Logging

409. How do transactions impact performance?
410. What's the difference between implicit and explicit transactions?
411. What is auto-commit mode?

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412. What's the impact of long-running transactions?
413. How do you detect long-running transactions?
414. How do you monitor open transactions?
415. What is the role of transaction log in performance?
416. How do you optimize log file growth?
417. What is minimal logging in bulk operations?
418. How do you achieve minimal logging?
419. How do you troubleshoot high log generation rate?
420. What is the impact of recovery model on performance?
421. What's the difference between full, bulk-logged, and simple recovery?
422. How do you minimize log backup impact on performance?
423. How do you measure log flushes/sec?
424. How do you monitor log send queues in AG?
425. How do you shrink transaction log safely?
426. What are ghost records?
427. How do you clean up ghost records?
428. How do you troubleshoot log file IO issues?

Maintenance & Performance

429. How do you schedule index maintenance jobs?
430. What's the impact of index rebuilds on performance?
431. How do you decide between online and offline index rebuilds?
432. What's the performance impact of updating statistics?
433. How often should you update statistics?
434. How do you balance index rebuilds with statistics updates?
435. How do you optimize DBCC CHECKDB performance?
436. How do you run CHECKDB on large databases?
437. What's the impact of backup compression on performance?
438. How do you monitor backup throughput?
439. How do you optimize full backups for VLDBs?
440. How do you tune differential backups?
441. How do you reduce log backup impact on performance?
442. How do you schedule maintenance jobs to reduce contention?
443. What's the impact of auto-shrink on performance?
444. How do you disable auto-shrink?

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445. How do you monitor job execution performance?
446. How do you troubleshoot failed maintenance jobs?
447. How do you automate performance baselining?
448. What are Ola Hallengren's scripts and why are they popular?

Advanced Tuning Topics

449. How do you optimize partitioned views?
450. How do you troubleshoot CROSS DATABASE queries?
451. How do you optimize queries against linked servers?
452. How do you reduce network latency in distributed queries?
453. How do you tune Service Broker performance?
454. How do you monitor Resource Governor workload groups?
455. How do you set CPU/memory caps for workloads?
456. How do you troubleshoot query timeouts?
457. What is query governor cost limit?
458. How do you tune filetable performance?
459. How do you optimize CLR functions?
460. How do you tune spatial data queries?
461. How do you optimize JSON functions?
462. How do you optimize XML shredding queries?
463. What's the performance impact of OPENJSON?
464. How do you optimize OPENROWSET queries?
465. How do you tune PolyBase queries?
466. How do you monitor external tables performance?
467. What is Adaptive Query Processing?
468. How do you use Adaptive Joins?

Real-Time Troubleshooting

469. How do you capture live blocking in SQL Server?
470. How do you capture queries with high CPU in real time?
471. How do you detect queries spilling to TempDB?
472. How do you troubleshoot SOS_SCHEDULER_YIELD waits?
473. How do you troubleshoot LCK_M_SCH waits?
474. How do you detect latch contention in TempDB?
475. How do you troubleshoot PAGELATCH_SH waits?

476. How do you detect excessive recompiles in queries?
477. How do you troubleshoot RESOURCE_SEMAPHORE waits?
478. How do you detect high compilation rate?
479. How do you troubleshoot parameter sniffing?
480. How do you detect skewed parallelism?
481. How do you troubleshoot excessive network IO?
482. How do you measure query latency end-to-end?
483. How do you analyze plan regressions after patching?
484. How do you troubleshoot excessive recompiles due to temp tables?
485. How do you monitor deadlocks proactively?
486. How do you troubleshoot blocking caused by statistics updates?
487. How do you fix slow inserts into heap tables?
488. How do you troubleshoot ghost record cleanup issues?

Interview-Style Scenario Questions

489. A query runs fast with one parameter but slow with another — what's wrong?
490. CPU usage is 100% but IO is normal — what do you check?
491. Queries are fast in dev but slow in prod — what do you compare?
492. A table has too many indexes — what's the impact?
493. Query uses an index scan instead of seek — why?
494. You see high PAGEIOLATCH waits — what's the root cause?
495. Blocking is reported during peak hours — how do you fix it?
496. After index rebuild, query performance worsened — why?
497. A plan changed after stats update — how do you force old plan?
498. TempDB is growing rapidly — what do you check?
499. Query Store shows multiple forced plans failing — what's your approach?
500. How do you design a performance tuning checklist for SQL Server?

Top 50 SQL Performance Tuning Questions:

1. What are the main causes of SQL Server performance issues?
2. How do you identify slow-running queries in SQL Server?
3. What DMVs are commonly used for performance troubleshooting?
4. What are clustered and non-clustered indexes, and how do they impact performance?
5. What is a covering index?
6. How do you detect missing indexes?
7. How do you avoid over-indexing a table?
8. What is fragmentation, and how do you fix it?
9. Difference between index rebuild and index reorganize?
10. What is parameter sniffing?
11. How do you resolve parameter sniffing issues?
12. How do you read and interpret an execution plan?
13. Difference between an index seek and an index scan?
14. How do you detect implicit conversions in a query?
15. What are common SQL Server wait types you look at first?
16. How do you troubleshoot high CPU utilization in SQL Server?
17. How do you troubleshoot high IO waits?
18. How do you troubleshoot memory pressure in SQL Server?
19. What are blocking and deadlocks? How do you detect and fix them?
20. How do you capture and analyze deadlock graphs?
21. What is TempDB, and why does it cause performance bottlenecks?
22. How do you optimize TempDB configuration?
23. What is the difference between a heap and a clustered table?
24. How do statistics affect query performance?
25. How often should you update statistics?
26. What's the difference between auto update statistics and manual update?
27. How do you use Query Store for performance troubleshooting?
28. How do you identify and fix query plan regressions?
29. What are common causes of recompilation in SQL Server?
30. What's the difference between adhoc query and stored procedure in terms of performance?
31. How do you monitor query performance in real time?
32. How do you optimize joins (nested loop, merge, hash)?
33. How do you choose between CTEs, views, and temp tables for performance?
34. What is the impact of cursors on performance, and how can they be avoided?

35. What is MAXDOP, and how does it affect parallelism?
36. What is cost threshold for parallelism, and how do you tune it?
37. How do you troubleshoot CXPACKET and CXCONSUMER waits?
38. How do you detect and resolve excessive recompiles?
39. What is page life expectancy (PLE), and what does it tell you?
40. What is buffer cache hit ratio?
41. How do you measure and improve query execution time?
42. What is minimal logging, and when can you achieve it?
43. How do you shrink the transaction log safely?
44. What are ghost records, and how do they affect performance?
45. How do you troubleshoot TempDB contention (PAGELATCH waits)?
46. How do you baseline SQL Server performance?
47. What tools do you use for SQL Server performance monitoring (native + third-party)?
48. What is Adaptive Query Processing?
49. Give a real-time example where statistics caused a query to run slow.
50. If a query runs fine in dev but slow in prod, what steps do you take to troubleshoot?

Top 10 Questions

1. What are the main causes of SQL Server performance issues?
2. How do you identify slow-running queries in SQL Server?
3. What are clustered and non-clustered indexes, and how do they impact performance?
4. What is parameter sniffing, and how do you resolve it?
5. How do you read and interpret an execution plan (seek vs scan, joins, operators)?
6. How do statistics affect query performance, and how often should they be updated?
7. What are blocking and deadlocks, and how do you detect and fix them?
8. What is TempDB, why does it cause performance bottlenecks, and how do you optimize it?
9. What is MAXDOP, cost threshold for parallelism, and how do you tune them?
10. If a query runs fine in dev but slow in prod, what steps do you take to troubleshoot?